Questions (12 marks)

1. Explain the main differences between RDDs, Dataframes and Datasets (4 marks)

   Resilient Distributed Dataset (RDD):

   - RDD is a collection of data without a schema.
   - The most basic of all 3
   - Developer needs to write any code and define the schema manually.
   - Simple operations like parsing, grouping, slicing, and sorting data are not ideal for RDD.
   - Ideal for unstructured data
   - Not columnar data like those found in Dataframes and Datasets.
   - More control when performing low level transformation and allows for manipulation of data with functional programming constructs.

   Dataframes:

   - Data distributed into organized named columns.
   - More functionality than an RDD
   - Optimization occurs using the catalyst optimizer.
   - Simple operations like parsing, grouping, and sorting data are great with Dataframes.
   - Ideal for structured data utilizing functions such as rich semantics, high-level abstractions, and domain specific APIs.
   - R and Python use Dataframes.
   - API is easy to use.

   Datasets:

   - Data distributed into organized named columns.
   - More functionality than an RDD
   - A more sophisticated collection than Dataframes with more features like runtime type-safety which will allows a syntax error to be discovered during compile time rather than runtime.
   - Optimization occurs using the catalyst optimizer.
   - Complex operations like parsing, grouping, and sorting data are great with Datasets.
   - Ideal for structured data
   - Dataset takes on two distinct APIs characteristics: a strongly-typed API and an untyped API
   - Tungsten Encoders can efficiently serialize/deserialize JVM objects allowing a higher level of efficiency during code generation.

2. Answer the following questions (**PROOF LOCATED ON GITHUB IPYNB FILE**):

   2.1 How many sensor pads are reported to be from Poland (2 marks)

   1413 sensor pads are reported from Poland.

2.2 How many different LCDs are present in the dataset (2 marks)?

3 Different Types: Green (49699), Yellow (99051) and Red (49414)

2.3 Find 5 countries that have the largest number of MAC devices used (2 marks)

Top 5 Countries

United States: 11509

China: 2300

Japan: 2002

Republic of Korea: 1999

Germany: 1314

2.4 Propose and try an interesting statistical test or machine learning model you could use to gain insight from this dataset (2 marks)

It would be interesting to see the battery level of devices used in less developed countries versus those found in the developed nations. Monitoring to see which devices would be considered optimal when analyzing humidity, temperature, battery levels and CO2 would be indicative of which countries are utilizing better devices.

My proof is on my personal github at https://github.com/ketankhanna91/Ketan-Files/blob/main/Assignment%203%20Python%20-%202021-06-20.ipynb